Convergence of gradient Descent for Convex Functions

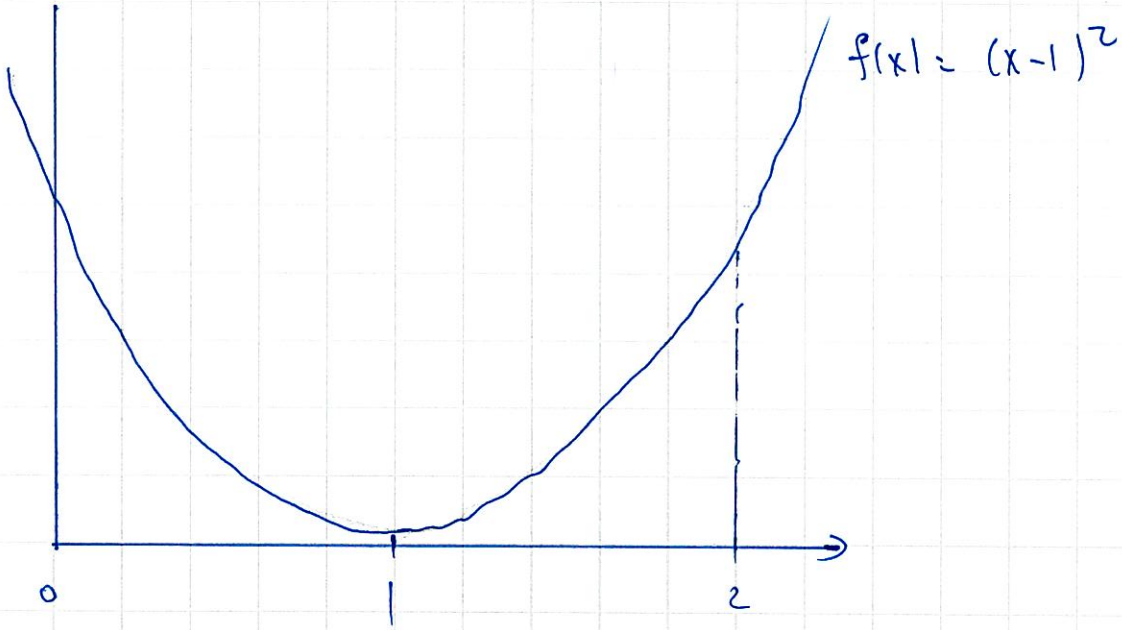
- We want to minimize a convex function $f : \mathbb{R}^n \to \mathbb{R}$ by using gradient descent with constant stepsize γ.

- We generate a sequence of points $x^0, x^1, x^2, \ldots$ by using

$$x^{k+1} = x^k - \gamma \nabla f(x^k)$$

where x^0 is the initial point we pick.

- With a constant step size, we will not be able to establish convergence to a minimum. The best we can accomplish is that we will bounce around a minimum. Even so, our convergence result will be a meaningful result.



$f'(x) = 2.(x-1)$

- Minimize $f(x) = (x-1)^2$ starting with $x^0 = 2$ with constant step size $\gamma = 1$

$f'(x^0) = f'(2) = 2$ $\quad$ $x^1 = x^0 - \gamma f'(x^0) = 2 - 1.2 = 0$

$f'(x^1) = f'(0) = -2$ $\quad$ $x^2 = x^1 - \gamma f'(x^1) = 0 - 1(-2) = 2$

$f'(x^2) = f'(2) = 2$ $\quad$ $x^3 = 0$ $\quad$ $x^4 = 2$ $\quad$ $x^5 = 0$ ...

So we will be bouncing around the minimum $x^* = 1$.

Preliminary Analysis

- Recall $f: \mathbb{R}^n \to \mathbb{R}$ is convex iff $f(y) \geq f(x) + \nabla f(x)^t \cdot (y-x)$ for all $x, y \in \mathbb{R}^n$

- Another useful equality is: $\|x-y\|^2 = \|x\|^2 + \|y\|^2 - 2x^t y$ (‡)

Pf: Remember $\|z\|^2 = z^t z$

$$\|x-y\|^2 = (x-y)^t (x-y) = (x^t - y^t)(x-y) = x^t x - x^t y - y^t x + y^t y$$
$$= \|x\|^2 - 2x^t y + \|y\|^2$$

- Here is the preliminary analysis we will use in our convergence proof

We have $x^{k+1} = x^k - \gamma \underbrace{\nabla f(x^k)}_{g^k}$

- For simplicity of notation, set $g^k = \nabla f(x^k)$, so $x^{k+1} = x^k - \gamma \cdot g^k$

- Let $x^* = \arg\min_{x \in \mathbb{R}^n} f(x) \leftarrow x^*$ is the minimum of f.

We want to make sure that we get close to the value of f at x^*.

- We want to upper bound $f(x^k) - f(x^*)$
- Let's start with first order characterization of convexity:

$$f(x^*) \geq f(x^k) + \nabla^t f(x^k) \cdot (x^* - x^k) \Rightarrow$$

$$f(x^k) - f(x^*) \leq \nabla^t f(x^k)\,(x^k - x^*) \Rightarrow f(x^k) - f(x^*) \leq (g^k)^t\,(x^k - x^*) \quad (*)$$

- Thus, to upper bound $f(x^k) - f(x^*)$, it is enough to upper bound $(g^k)^t\,(x^k - x^*)$
- From gradient descent $x^{k+1} = x^k - \gamma g^k \Rightarrow g^k = \frac{1}{\gamma} \cdot (x^k - x^{k+1})$
- Let's go ahead and upper bound $(g^k)^t\,(x^k - x^*)$.

- $$(g^k)^t (x^k - x^*) = \frac{1}{\gamma} \cdot (x^k - x^{k+1})^t \cdot (x^k - x^*)$$

$$\text{from } \mathcal{F} \quad = \frac{1}{2\gamma} \left(\|x^k - x^{k+1}\|^2 + \|x^k - x^*\|^2 - \|(\cancel{x^k} - x^{k+1}) - (\cancel{x^k} - x^*)\|^2 \right)$$

$$= \frac{1}{2\gamma} \left(\gamma^2 \|g^k\|^2 + \|x^k - x^*\|^2 - \|x^{k+1} - x^*\|^2 \right)$$

$$= \frac{\gamma}{2} \|g^k\|^2 + \frac{1}{2\gamma} \cdot \left(\|x^k - x^*\|^2 - \|x^{k+1} - x^*\|^2 \right)$$

- Add the inequality above over $k = 0, 1, 2, \ldots, K-1$

$$\sum_{k=0}^{K-1} (g^k)^t (x^k - x^*) = \frac{\gamma}{2} \cdot \sum_{k=0}^{K-1} \|g^k\|^2 + \frac{1}{2\gamma} \left(\|x^0 - x^*\|^2 - \|x^K - x^*\|^2 \right)$$

(By telescoping sums)

$$\leq \frac{\gamma}{2} \sum_{k=0}^{K-1} \|g^k\|^2 + \frac{1}{2\gamma} \|x^0 - x^*\|^2$$

- Using $(*)$, we get

$$\sum_{k=0}^{K-1} \left(f(x^k) - f(x^*) \right) \leq \sum_{k=0}^{K-1} (g^k)^t (x^k - x^*) \leq \frac{\gamma}{2} \sum_{k=0}^{K-1} \|g^k\|^2 + \frac{1}{2\gamma} \|x^0 - x^*\|^2$$

End of preliminary analysis.

- out line of preliminary analysis
  - Start with first-order characterization of convexity
  - Use $\|x-y\|^2 = \|x\|^2 + \|y\|^2 - 2x^t y$ to convert scalar product into norm difference
  - Use a telescoping sum.

- The last inequality has some meaning.
  - $\sum_{k=0}^{K-1} (f(x^k) - f(x^*))$ is the cumulative "error" over the first K iterations.
  - If γ is small, the upper bound in the error is dominated by $\frac{1}{2\gamma}\|x^0 - x^*\|^2$.

    If stepsize is small, we will get stuck close to the initial point x^0. so the error will be dominated by how close the initial point is to x^*

  - If γ is large, then the upper bound on the error is dominated by

$$\frac{\gamma}{2}\sum_{k=0}^{K-1}\|g^k\|^2.$$

If the stepsize is large, whenever the gradient is non-zero, we keep on overshooting.

Convergence of gradient Descent under Bounded gradients:

- Assume f is convex and $\|\nabla f(x)\| \leq B$ for all $x \in \mathbb{R}^n$

Theorem: Let f be convex with minimum x^*. Assume that $\|\nabla f(x)\| \leq B$ for all x and we start gradient descent at point x^0 with $\|x^0 - x^*\| = R$

If we choose the stepsize as $\gamma = R/B\sqrt{K}$, then the iterates of gradient descent satisfy

$$\frac{1}{K} \sum_{k=0}^{K-1} \left(f(x^k) - f(x^*) \right) \leq \frac{RB}{\sqrt{K}}$$

So the average "error" of gradient descent diminishes with rate $1/\sqrt{K}$ after K iterations.

Pf: From our preliminary analysis:

$$\sum_{k=0}^{K-1} \left(f(x^k) - f(x^*)\right) \leq \frac{\gamma}{2} \sum_{k=0}^{K-1} \|g^k\|^2 + \frac{1}{2\gamma} \|x^0 - x^*\|^2$$

$$\leq \frac{\gamma}{2} K \cdot B^2 + \frac{1}{2\gamma} R^2$$

- So the cumulative error over K iterations is upper bounded by $\frac{\gamma}{2} K B^2 + \frac{1}{2\gamma} R^2$

Choose γ to make this upper bound as small as possible.

$$H(\gamma) = \frac{1}{2} \cdot \left(K B^2 \gamma + R^2 \frac{1}{\gamma}\right) \qquad H'(\gamma) = \frac{1}{2} \cdot \left(K B^2 - R^2 \frac{1}{\gamma^2}\right) = 0 \Rightarrow$$

$$\gamma = \sqrt{\frac{R^2}{K B^2}} \Rightarrow \gamma = \frac{R}{B\sqrt{K}}$$

- With this stepsize: $$\sum_{k=0}^{K-1} \left(f(x^k) - f(x^*)\right) \leq \frac{1}{2} \cdot \left(\frac{R}{\cancel{B}\cancel{\sqrt{K}}} \cdot \cancel{K}^{\sqrt{K}} \cdot B^{\cancel{2}} + \frac{B\sqrt{K}}{\cancel{K}} R^{\cancel{2}}\right)$$

$$= R B \sqrt{K}$$

- Thus, $$\frac{1}{K} \sum_{k=0}^{K-1} \left(f(x^k) - f(x^*)\right) \leq \frac{1}{\sqrt{K}} R B$$ done!